

# Cancer Clinical Trial Curation

*Using Agentic Workflow*

**Hartwig Medical Foundation, Australia**  
**30 July 2026**





# The scale and the rate of change require automation

Hartwig curates the trial universe for Australia and New Zealand so that a patient's molecular profile can be matched against the trials open to them. Two properties of that universe determine how it has to be built.

## SCALE

### 2,041 open oncology trials

Across two registries: **5,284** trial arms and **17,830** distinct combinations of eligibility requirements.

The eligibility of each arm, and the drugs it administers, are described in **free text**.

## CONTINUOUS CHANGE

### Records are added, amended and closed

Between our **25 July** snapshot and the **29 July** download — four days — **71** trials had entered the universe and **32** had left it.

A curated resource therefore has a short shelf life.

At this scale and this rate of change, **both ingestion and curation have to be automated** and repeatable, so that the resource can be rebuilt on a schedule.





# From free text to a structured, matchable resource

The key sections of each trial arrive as free text. We transform them into a **structured resource file with controlled vocabulary** for the downstream matching engine. It has three sections.

## SECTION 1

### Basic details

Recruitment status, phase, sponsor, locations — the basic info fields, carried through deterministically.

## SECTION 2

### Eligibility

Who qualifies — cancer type, gene alterations, biomarkers, prior therapy — as structured logic in controlled vocabulary (OncoTree, finding-model syntax).

## SECTION 3

### Drug utility

The interventional drugs — canonical identity, class, targets, and TGA / PBS approval status.

All three sections must be **machine-interpretable**: every value in controlled vocabulary where one exists, and every value traceable back to the source text it came from.





# Transforming free text into structured vocabulary is difficult

Eligibility criteria are written for human readers. The structure a matching engine needs is present in the text but not marked, and has to be recovered before any value can be mapped to vocabulary. Four recurring difficulties:

## DIFFICULTY 1

### Alternatives are not delimited

“with MYCN amplification, or focal high level MYC or MYCL amplification” `NCT04221035` states **three** alternatives, of which a patient needs one. Nothing in the text marks where one ends and the next begins.

## DIFFICULTY 2

### Requirements combine

A cancer type, an alteration and a prior-therapy condition usually apply together, and different arms of the same trial combine them differently. Each valid combination is a separate way to qualify.

## DIFFICULTY 3

### Exclusions have a scope

An exclusion may apply to the whole trial or to one arm of it. “not eligible for the surgical cohort” constrains that cohort only, and applying it trial-wide would exclude eligible patients.

## DIFFICULTY 4

### Terminology varies

The same disease appears as an abbreviation, a full name, or a list of histologies. Across the trial universe there are **4,971** distinct ways of writing a cancer type, and each has to resolve to the correct code.





# Where our first attempt fell short

Version 1 was a linear chain of one-shot LLM calls over the eligibility section, corrected manually afterwards. Five problems motivated the rewrite.

## PROBLEM 1

### No automated verification

A linear chain of one-shot LLM calls requires a human reviewer. Virtually impossible to manually detect all errors across ~2,000 trials.

## PROBLEM 2

### No combinations of eligibility criteria

Cancer type and gene alterations were treated as independent criteria, so cancer × gene combinations needed manual overwriting.

## PROBLEM 3

### No external web search for drug enrichment

Drug annotations — drug class, target–mechanism, and **TGA / PBS** status — cannot be filled from the protocol text alone.

## PROBLEM 4

### Reliance on handcrafted mapping files

No reliable means of transforming input text into our controlled vocabulary without manually created mapping files.

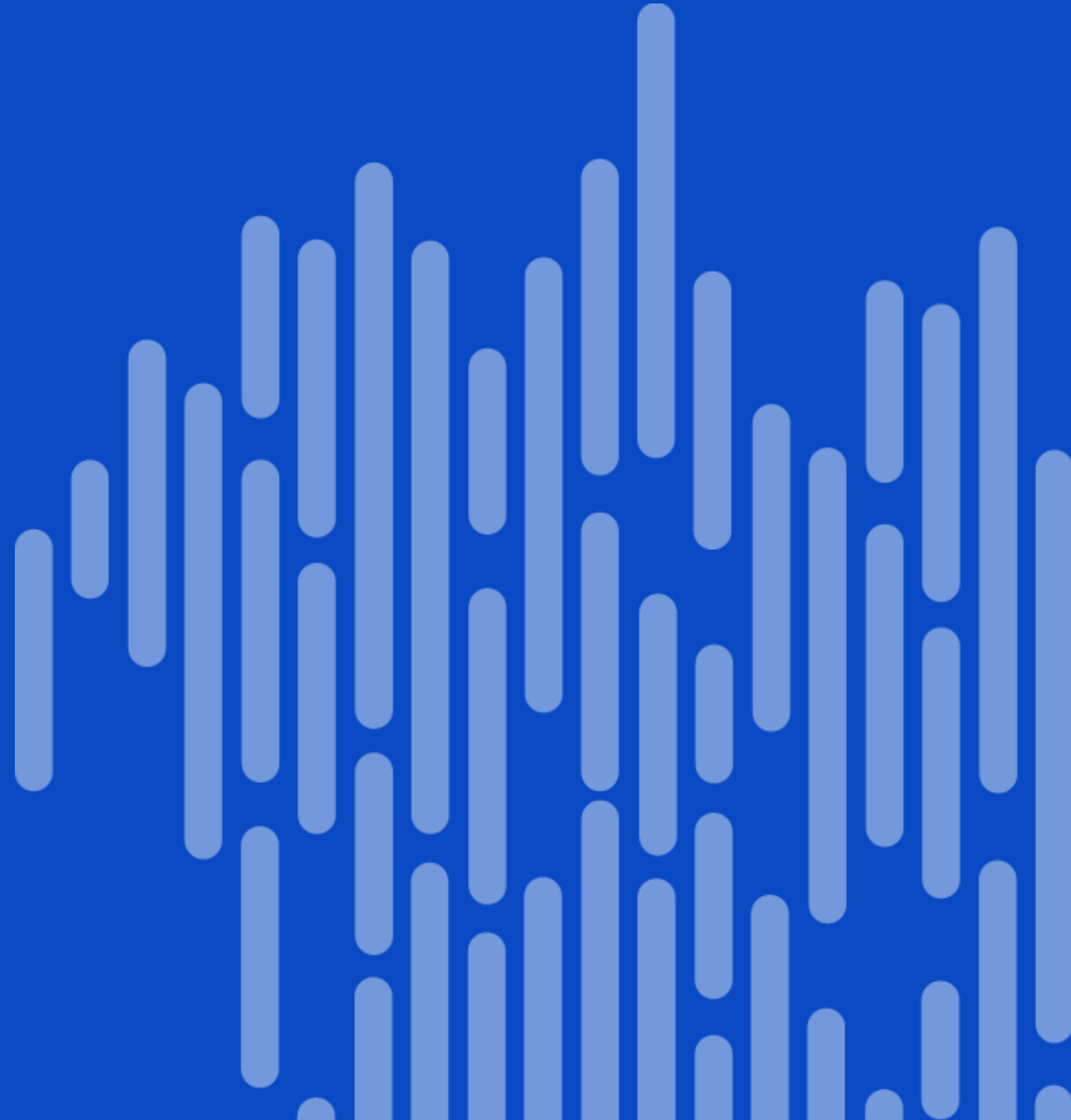
## PROBLEM 5

### Read only the eligibility section

Eligibility-relevant detail also appears in the **title, description, conditions and arm text**, which version 1 did not read.



# The pipeline, end to end



# The whole pipeline at a glance

## PART 1 · TRIAL INGESTION AND UPDATING

downloaded **in full** · kept if **open to recruitment, AU / NZ site, drug intervention, cancer condition**

### Assemble the source, and identify the arms of the trial

**CODE** / **DOER** → **REVIEWER** ↺

every section kept with its origin — **TITLE** **CONDITIONS** **DESCRIPTION** **ELIGIBILITY** **ARMS** . ClinicalTrials.gov lists its arms; ANZCTR does not.

## PART 2 · TRIAL CURATION – TWO INDEPENDENT PATHS

### ELIGIBILITY – FOR EVERY ARM

Quote each criterion, verbatim

**DOER** → **REVIEWER** ↺

Turn the prose into logic

**DOER** → **REVIEWER** ↺

Map to controlled vocabulary

**DOER** → **REVIEWER** ↺

Reconcile across all trials

**CODE**

### DRUG UTILITY – FOR EVERY DRUG, ONCE

Resolve the drug's identity

**DOER** → **REVIEWER** ↺

Find its mechanism and class

**DOER** → **REVIEWER** ↺

Find its TGA / PBS status

**DOER** → **REVIEWER** ↺

Label its role in the arm

**DOER** → **REVIEWER** ↺

### Join both paths on the arm, and write one flat file

**CODE**

the matching engine reads this file — one row per set of requirements a patient can satisfy

**code** — no LLM involved   **doer** — the LLM that writes the answer   **reviewer** — an LLM that checks it   ↺ sent back





# Where the trials come from

Two registries, downloaded in full on every run and filtered locally to the trials we curate.

CLINICALTRIALS.GOV — 1,525 TRIALS

## Retrieved by API

A paged search with the filters applied at the registry.

ANZCTR — 516 TRIALS

## Retrieved as a whole-registry export

ANZCTR has no search API, so we download the entire registry (91 MB) and apply the same filters locally. **777** trials matched; **516** had a drug intervention.

A PRIORITY LIST THAT OVERRIDES THE FILTERS

A published list of Australian trials must always be included. It widens the registry search and exempts those trials from the drug-intervention filter — **30** non-drug trials were kept on that basis.

VERSIONED INPUTS

Each download becomes the current version and the previous one is archived, so a run can be reproduced against the exact input it read.





# Keeping the resource up to date

The pipeline is designed to be re-run regularly. Each cycle compares the fresh download against what is already curated, and only work that has not been done before consumes any LLM calls.

1 · RE-DOWNLOAD BOTH REGISTRIES IN FULL – apply the filters → the current set of trials we keep

2 · each trial falls into one of three cases. “New” here means **new to the kept set**, which is not the same as newly registered

IN THE DOWNLOAD, NOT YET CURATED

## New → curate it

Either a new registration, or an **existing trial whose record changed** so that it now matches the filters. Within it, only drugs absent from the reference are researched.

71 trials · 25 → 29 July

IN THE DOWNLOAD, ALREADY CURATED

## Unchanged → skip it

The stored answer is returned against a fingerprint of the request, so no LLM call is made.  
~1,960 trials, near-instant

CURATED, ABSENT FROM THE DOWNLOAD

## Retired → moved aside

Its recruitment status changed — completed, withdrawn or terminated — so it no longer matches. Its rows are **moved** to a holding area, and **restored intact** if it returns.

32 trials · 25 → 29 July

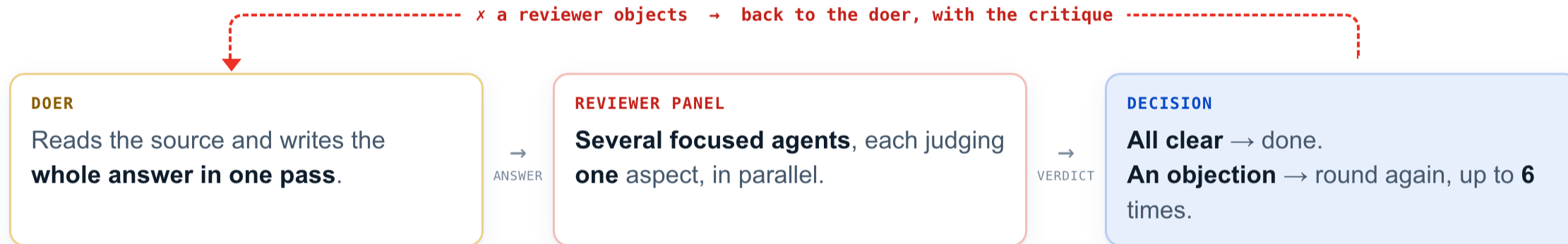
3 · the common case, from this week. `NCT06652438` is a phase 3 AML trial first submitted on **2 October 2024**, with no Australian site. Its record was updated on **27 July 2026** and posted on **29 July**, adding **10 Australian sites** — so it began matching the location filter and entered the kept set that day. An international trial opening Australian sites is precisely the change a regular refresh exists to catch.





## Both curation paths use the same pattern: a **doer** and its **reviewers**

Every LLM step on the eligibility path and the drug utility path has the same shape: one agent writes the whole answer, focused reviewers judge it against the source, and any objection sends it back to be rewritten.



### WHY CHECKING WORKS BETTER THAN WRITING

Producing a correct answer is a search problem; **checking a proposed one is closer to a look-up**. Think of a mathematical proof: finding it can take years, while checking a written proof is a careful read. So we spend the most capable effort on the doer, and can afford **several** cheaper reviewers to catch its mistakes.

### THE EXTRACTION PANEL – FIVE REVIEWERS, ONE ASPECT EACH

A narrow check is far more reliable than one agent judging everything at once.

**cancer type** **molecular** **structure** **prior therapy**  
**drug**

A reviewer can object but **never rewrites** — so every value in the output was **written by the doer and then checked**.



# What code computes, and what an LLM must judge

Anything with one correct value that can be looked up is resolved by code, and is never asked of an LLM.

## COMPUTED PROGRAMMATICALLY

### Facts with one right answer

- drug identifiers and class codes, from a drug reference
- whether a code or an expression is **valid** at all
- which trials are new and which have gone
- every join, de-duplication and the final file

The LLM supplies a drug's **name**; code turns that name into its **identifier**. A wrong identifier would corrupt every join that depends on it.

## JUDGED BY AN LLM

### Reading, interpretation, search

- what a sentence actually **requires** of a patient
- which alternatives are real, and which are examples
- which drug an arm is **testing**, and which is supportive
- finding a drug's mechanism and approval status **on the web**
- whether a mapping is **faithful** to the source text

Each of these requires reading the source and forming a judgement about it.

Each side is given only the work it is reliable at: **the LLM reads and interprets; the code computes, validates and joins.**





# Spot checks — every individual value, as it is produced

Each value is checked twice before it is stored: once for **faithfulness** to the source, and once for **form**.

## IS IT FAITHFUL TO THE SOURCE — CHECKED BY AN LLM REVIEWER

- Does every quoted span really appear in the source
- Is the cancer type the tumour **under study**
- Is each molecular value in the **right column**
- Are only **required** and **excluded** prior therapies treated as constraints
- Is the row structure and per-arm scope correct
- Is the vocabulary code faithful, at the right level of detail
- For a drug: the right ingredients, a plausible mechanism and class, sourced TGA / PBS entries, and its role in the arm

## IS IT WELL-FORMED — CHECKED BY CODE

- Is the cancer code **valid**, and the alteration expression **syntactically valid**
- Is anything **self-contradictory** — the same term required and excluded
- Is a code **duplicated**, or a broad type ANDed with its own subtype
- Is the **OR** / **AND** precedence unambiguous
- Did the drug identifier, class and ATC code come from a **lookup**
- Are exact-duplicate and merely re-ordered rows removed





# Overall checks — the finished cycle, before it is trusted

Spot checks cannot detect a cycle that curated only half the universe and still produced a plausible-looking file. **Ten deterministic gates** examine the finished state on disk — no LLM, no network.

## integrity fail

Every eligibility, drug and role row points at an arm that exists.

## retirement fail

A retired trial is gone from **every** table.

## nothing lost fail

Reference data never shrank, and  
`after = before - retired + newly curated` holds exactly.

## completeness fail

Every trial in the fresh universe is in the store.

## output validator warn

An independent pass re-checks the finished file and reports how many trials are clean.

## empty output fail

Every arm producing **no** eligibility carries a reason. An unexplained empty is a possible miss.

## export fail

Exactly the expected columns; no curated trial missing without a reason.

## download sanity fail

Retirement was not skipped by its own safety guard.

## universe swing warn

The kept universe changed by less than **10%** against the previous cycle.

## waived findings warn

Known, accepted findings stay visible as a warning, so a gate fails on **new** problems rather than on a standing backlog. Each waiver is code, reviewed in git, and names its follow-up.

### HOW A FAILURE SURFACES

Any **FAIL** makes the run **exit non-zero** so the scheduler sees it, and writes `STATUS.json` for a monitor to poll. **WARN** means look, but the data is usable.

### A DURABLE RECORD PER CYCLE

One Markdown **run report**: the universe, **which trial ids** were retired, restored and newly curated, a before–after delta for every table, and each gate's verdict.





# Three real corrections, one from each layer of checking

## CAUGHT BY AN LLM REVIEWER

### An example became a requirement

The source excluded EGFR mutations, then added “*gene mutations that do not routinely use immunotherapy (e.g. ALK fusion)*”.

The doer made the example **mandatory**:

NOT(EGFR) AND Fusion[ALK]

The reviewer objected: the phrase is open-ended and ALK is only an illustration — requiring it **narrows the trial to patients it never specified**.

Corrected to drop the term.

## CAUGHT BY A DETERMINISTIC CHECK

### A mapping that contradicted itself

A mapping came back with the same cancer code both **required** and **excluded** in one expression:

... PCM ... AND NOT(PCM)

No patient can satisfy that, so the row would silently match nobody. The validator rejects **X AND NOT(X)** outright — it needs no judgement, so it never gets one wrong. The usual cause: two cohorts' requirements folded into one row.

## CAUGHT ACROSS ALL TRIALS

### The same disease, two different codes

Mapped trial by trial, equivalent phrasings drift apart. A run-wide pass compares every value across all trials: code normalises the equivalent ones, and an LLM adjudicates genuine conflicts.

“Astrocytoma”: DIFG → ASTR

“recurrent lymphoma”: HL OR NHL → LNM

**185 of 4,971** cancer-type values were changed by this pass — each one a trial that would otherwise have been unmatchable against its own disease.

The three layers detect **different classes of error**, and each of these corrections was found by only one of them.



# What it produces, and what we learned





# One trial, end to end — NCT07247110

Three arms, each a different drug regimen. Two accept **any solid tumour** after prior therapy; the third accepts **only lung cancer**, in **untreated** patients.

**Dose escalation and + Cetuximab:** “locally advanced unresectable or metastatic solid tumor” · “at least 1 prior line of systemic therapy”  
**+ Pembrolizumab:** “metastatic non-small cell lung cancer” · “must be untreated” · “has received any prior immunotherapy” (excluded)  
**All three:** “presence of Kirsten rat sarcoma viral oncogene homolog (KRAS) alteration”  
quoted verbatim from each arm's eligibility criteria

| ARM — DRUG REGIMEN | CANCER TYPE                           | → CODE       | KRAS ALTERATION → MATCHABLE                                                                | PRIOR THERAPY                                  |
|--------------------|---------------------------------------|--------------|--------------------------------------------------------------------------------------------|------------------------------------------------|
| MK-4716 alone      | advanced / metastatic solid tumour    | Solid tumour |                                                                                            | ≥1 prior line of therapy                       |
| + Pembrolizumab    | metastatic non-small cell lung cancer | NSCLC        | SmallVariant[ <i>gene</i> =KRAS]<br>  GainDeletion[ <i>gene</i> =KRAS & <i>type</i> =GAIN] | untreated <b>AND NOT</b> (prior immunotherapy) |
| + Cetuximab        | advanced / metastatic solid tumour    | Solid tumour |                                                                                            | ≥1 prior line of therapy                       |

## WHAT THIS ACHIEVES

Each arm retains **its own** cancer type and prior-therapy rule. Version 1 produced a single list of criteria per trial, under which a previously-treated patient would appear eligible for the pembrolizumab arm. The gene name, written out in full, resolves to **two precise terms**, so a KRAS point mutation and a KRAS amplification both match.





WORKED EXAMPLE

## One drug, end to end — pembrolizumab

A drug's facts are properties of the drug itself, so each is researched **once** and reused across every arm that administers it. **2,899** raw intervention names resolve to **1,310** distinct drugs.

### STEP 1 · IDENTITY

#### Brand and code names, resolved

“Keytruda”, “MK-3475” → **pembrolizumab**. The LLM supplies the name; the identifier `rxcul:1547545` comes from a **lookup**.

### STEP 2 · WHAT IT IS AND DOES

#### Mechanism, by web search

Monoclonal antibody — target **PD-1**, action **inhibitor**. Its class codes are then attached by lookup.

### STEP 3 · REGULATORY STATUS

#### TGA and PBS, per indication

Web search of official sources, per approved indication, with the **date and an evidence link**. This information does not appear in the protocol text.

### STEP 4 · ROLE IN THE ARM

#### Tested, or supporting

The same drug is the **subject** of one trial arm and a **backbone or comparator** in another. This judgement is made per arm; everything above it is shared across arms.

### WHY THIS MATTERS FOR MATCHING

Approval is **indication-specific** — so we map each drug's approved indications into the **same** OncoTree and finding-model vocabulary as the trial side, and the two can be compared directly.





# Mapping prompts, developed with ML methodology

Mapping free text to a controlled vocabulary is where we borrow ML discipline. Our own **hand-curated resource files** are the ground truth, and are **held out** as the test set.

## THE DOER PROMPT

### Grammar and worked examples

The ontology and grammar, plus a handful of **positive examples** taken from the curated files — what a correct mapping looks like.

## THE REVIEWER PROMPT

### Counter-examples

Deliberately seeded with **plausible-looking wrong mappings** — the traps — so the reviewer learns what to reject, not only what to accept.

## THE PROCESS

### Train, then validate

Live-test on one set, fix at the **level of a rule** rather than a single case, then **validate on a disjoint set** to guard against overfitting.

| MAPPING                                       | TRAINED ON            | HELD-OUT VALIDATION    | PASSED REVIEW            |
|-----------------------------------------------|-----------------------|------------------------|--------------------------|
| <b>cancer type</b> → <b>OncoTree</b>          | 50 trials             | 100 trials, 288 values | <b>287 / 288 (99.7%)</b> |
| <b>gene alteration</b> → <b>finding-model</b> | 50 trials, 330 values | 100 trials, 242 values | <b>239 / 242 (98.8%)</b> |

The rest of each curated resource file is never shown to the mappers, so the validation set is genuinely unseen.





THE RESULT

# What the matching engine receives

2,023

trials in the deliverable

5,284

trial arms

17,830

eligibility paths

1,310

distinct drugs

8

linked drug tables

One flat file for eligibility, plus a set of linked drug tables. **One row = one combination of requirements a patient can satisfy** for one arm of one trial, so the matching engine compares values directly.

## EVERY VALUE STAYS TRACEABLE

Each interpreted value keeps the **quoted source text** it came from and the **section** it was found in, so any row in the output can be taken back to the sentence that produced it.

## RE-RUNS ARE CHEAP ON PURPOSE

Every LLM call is stored against a fingerprint of its exact request, so an unchanged trial costs **nothing** to re-process. Change a prompt and only the work that prompt affected is recomputed.





# Where it still struggles

## LIMIT 1

### The hardest trials do not converge

On deeply nested, many-cohort protocols the doer cannot satisfy every reviewer within the round budget. In our last full run, **219 of 1,999 trials (11%)** ended on a best-effort answer.

Those outputs are **written and marked** for human review, rather than dropped.

## LIMIT 2

### Correctness costs compute

Because a reviewer may object but never edit, the doer **rewrites the whole answer** each round rather than patching it. This costs more tokens and more time than a targeted edit.

## LIMIT 3

### Reviewers only catch what they are asked

A mistake that no reviewer is looking for will pass. We found one such gap and closed it by adding a reviewer that reads the whole table; we cannot demonstrate that no further gap remains.





● THE RESULT

# How long curation takes

The **71** trials newly curated in the 29 July cycle, measured end to end — quoting, interpreting, mapping and their drugs, including every review round.

**22<sub>s</sub>**

fastest trial

**1.9<sub>min</sub>**

median trial

**20.5<sub>min</sub>**

slowest trial

**38<sub>min</sub>**

all 71 trials, start to finish

## THE SPREAD

### Most trials are quick

**57 of the 71** finished in under five minutes. The long tail is made up of multi-cohort protocols that take several rounds with their reviewers.

## WHY THE TOTAL IS NOT THE SUM

### Trials are curated concurrently

The whole set took **38 minutes** rather than the four hours the individual times add up to.





# Thank you!

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